WORKED SOLUTION

STEP II 1987, Question 6

Pure • Definite Integrals by Symmetry

STEP II 1987, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s2-q6

The key idea

The interval $[-\frac{1}{2}\pi, \frac{1}{2}\pi]$ is symmetric, so the reflection $\theta \mapsto -\theta$ is the natural first move: it turns $\sin\theta$ into $-\sin\theta$ and gives the second form for free. Averaging the two forms removes the odd part and leaves a denominator in $\sin^2\theta$ only, which a little algebra splits into a constant plus a standard $\tan\theta$ integral. The final part is about one innocent-looking step: $\sqrt{\cos^2 2\alpha}$ is $|\cos 2\alpha|$, not $\cos 2\alpha$.

Solution

Throughout, write $s = \sin 2\alpha$ and $c = \cos 2\alpha$, so $s^2 + c^2 = 1$. For $0 < \alpha < \frac{1}{4}\pi$ we have $0 < s < 1$ and $0 < c < 1$. Since $s < 1$, the denominator $1 \mp s\sin\theta$ is positive on the whole interval, so all the integrals are proper.

The second form of I

Substitute $\theta = -\phi$, so $\mathrm{d}\theta = -\mathrm{d}\phi$ and the limits swap:

$$I = \int_{\frac{1}{2}\pi}^{-\frac{1}{2}\pi} \frac{\cos^2(-\phi)}{1 - \sin(-\phi)\, s}\,(-\mathrm{d}\phi) = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\phi}{1 + s\sin\phi}\,\mathrm{d}\phi,$$

using $\cos(-\phi) = \cos\phi$ and $\sin(-\phi) = -\sin\phi$. Renaming ϕ as θ gives the required form.

The expression involving $\cot^2 2\alpha$

Adding the two expressions for I,

$$2I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \cos^2\theta \left(\frac{1}{1 - s\sin\theta} + \frac{1}{1 + s\sin\theta}\right)\mathrm{d}\theta = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2\cos^2\theta}{1 - s^2\sin^2\theta}\,\mathrm{d}\theta,$$

so

$$I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\theta}{1 - s^2\sin^2\theta}\,\mathrm{d}\theta.$$

Now rewrite the denominator using $1 = \cos^2\theta + \sin^2\theta$ and $1 - s^2 = c^2$:

$$1 - s^2\sin^2\theta = \cos^2\theta + c^2\sin^2\theta.$$

Ask yourself How do we split $\dfrac{\cos^2\theta}{\cos^2\theta + c^2\sin^2\theta}$ into a constant plus something we can integrate?

Aim to write the numerator as a multiple of the denominator plus a remainder with no θ-dependence after dividing by $\cos^2\theta$. Since

$$(\cos^2\theta + c^2\sin^2\theta) - c^2(\cos^2\theta + \sin^2\theta) = (1 - c^2)\cos^2\theta = s^2\cos^2\theta,$$

we have, dividing by $s^2(\cos^2\theta + c^2\sin^2\theta)$ (allowed, as $s \neq 0$),

$$\frac{\cos^2\theta}{\cos^2\theta + c^2\sin^2\theta} = \frac{1}{s^2} - \frac{c^2}{s^2}\cdot\frac{1}{\cos^2\theta + c^2\sin^2\theta} = \frac{1}{s^2} - \frac{c^2}{s^2}\cdot\frac{\sec^2\theta}{1 + c^2\tan^2\theta},$$

where in the last step we divided top and bottom by $\cos^2\theta$. Integrating over an interval of length π, and using $c^2/s^2 = \cot^2 2\alpha$,

$$I = \frac{\pi}{\sin^2 2\alpha} - \cot^2 2\alpha \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\sec^2\theta}{1 + \cos^2 2\alpha\tan^2\theta}\,\mathrm{d}\theta,$$

as required.

Evaluating I

Call the remaining integral J. Substitute $t = \tan\theta$, so $\mathrm{d}t = \sec^2\theta\,\mathrm{d}\theta$; as θ runs from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, t runs over the whole real line. Then, for $c \neq 0$,

$$J = \int_{-\infty}^{\infty} \frac{\mathrm{d}t}{1 + c^2t^2} = \left[\frac{1}{|c|}\arctan(|c|\,t)\right]_{-\infty}^{\infty} = \frac{\pi}{|c|}.$$

(The substitution $u = |c|t$ is what produces $|c|$: it keeps the limits running from $-\infty$ to ∞ in the right order.)

For $0 < \alpha < \frac{1}{4}\pi$ we have $c > 0$, so $J = \pi/c$ and

$$I = \frac{\pi}{s^2} - \frac{c^2}{s^2}\cdot\frac{\pi}{c} = \frac{\pi(1-c)}{1-c^2} = \frac{\pi}{1+c} = \frac{\pi}{1+\cos 2\alpha} = \frac{\pi}{2\cos^2\alpha} = \tfrac{1}{2}\pi\sec^2\alpha,$$

cancelling $1 - c \neq 0$ and using $1 + \cos 2\alpha = 2\cos^2\alpha$. This is the required result.

The case $\frac{1}{4}\pi < \alpha < \frac{1}{2}\pi$

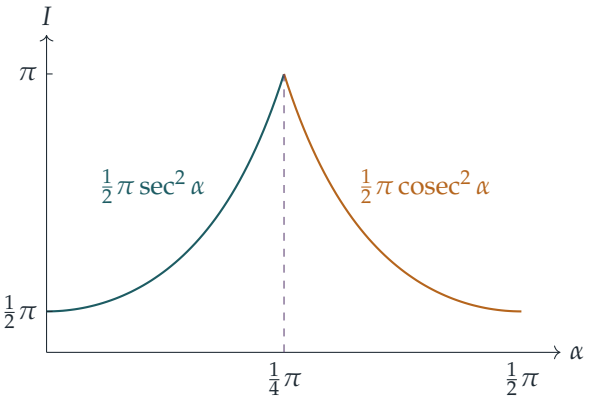
Ask yourself Does anything in the argument change when α passes $\frac{1}{4}\pi$?

Almost nothing: we still have $0 < s < 1$, so every step up to the formula for I in terms of J holds unchanged. The one difference is that now $c = \cos 2\alpha < 0$, so $|c| = -c$ and $J = -\pi/c$. Hence

$$I = \frac{\pi}{s^2} + \frac{c^2}{s^2}\cdot\frac{\pi}{c} = \frac{\pi(1+c)}{1-c^2} = \frac{\pi}{1-c} = \frac{\pi}{2\sin^2\alpha} = \tfrac{1}{2}\pi\operatorname{cosec}^2\alpha,$$

using $1 - \cos 2\alpha = 2\sin^2\alpha$. Blindly copying the first answer would give the wrong value.

The graph of I against α shows the two formulae meeting at $\alpha = \frac{1}{4}\pi$, where both equal π:



Checks

- At $\alpha = \frac{1}{4}\pi$ we have $s = 1$ and the integrand is $\cos^2\theta/(1-\sin\theta) = 1+\sin\theta$, which integrates to π. Both formulae give π there, as they should.
- At $\alpha = \frac{1}{6}\pi$: $s = \frac{\sqrt{3}}{2}$, $c = \frac{1}{2}$, so $I = \pi/(1+c) = \frac{2}{3}\pi$, and $\frac{1}{2}\pi\sec^2\frac{1}{6}\pi = \frac{1}{2}\pi\cdot\frac{4}{3} = \frac{2}{3}\pi$.
- Simpson's rule on the original integral at ten random values of α on both sides of $\frac{1}{4}\pi$ agrees with the formulae to better than 10^{-6}; for example $\alpha \approx 0.6308$ gives $I \approx 2.408759$ both ways.
- As $\alpha \to 0$ the integrand tends to $\cos^2\theta$, whose integral is $\frac{1}{2}\pi$, matching $\frac{1}{2}\pi\sec^2 0$.

Common slips

- Writing $\int_{-\infty}^{\infty}\frac{\mathrm{d}t}{1+c^2t^2} = \frac{\pi}{c}$ without the modulus. For $c < 0$ this is negative, which is impossible for a positive integrand.
- Losing the minus sign from $\mathrm{d}\theta = -\mathrm{d}\phi$, or forgetting that it is cancelled by swapping the limits.

- Worrying that $t = \tan\theta$ is undefined at $\theta = \pm\frac{1}{2}\pi$: the t-integral is improper but converges, which is all we need.

Beyond STEP

The tidy symmetry $\alpha \leftrightarrow \frac{1}{2}\pi - \alpha$ in the answer is no accident: I depends on α only through $\sin 2\alpha$, which is unchanged by that swap, while a formula like $\frac{1}{2}\pi\sec^2\alpha$ is not. Algebra that silently squares and square-roots can hide exactly this kind of issue. Integrals of rational functions of $\sin\theta$ and $\cos\theta$ over a full period (such as our averaged integrand, which has period π) are routinely done at university by contour integration around the unit circle, where the residue theorem makes them almost mechanical. For a beautifully geometric introduction, see Needham, *Visual Complex Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.